

Quiz 5 - D

Q1: Let A be the set of all strings of 0's, 1's, and 2's of length 4. Define a relation R on A as follows: For every $s, t \in A$,

$s R t \Leftrightarrow$ the sum of the characters in s equals the sum of the characters in t .

Is $1220 R 2111$? **(2 Marks)**

Q2: Determine whether the following relation is reflexive, symmetric, transitive:

F is the congruence modulo 5 relation on $\mathbb{Z}$ (set of all integers):

For every $m, n \in \mathbb{Z}$, $m F n \Leftrightarrow 5 \mid (m - n)$ **(4 Marks)**

Q3: A relation S on the set $A = \{1, 2, 3, 4\}$ is given by the matrix below (e.g. the value 0 in row 1, column 2 indicates that 1 is not related to 2, whereas the value 1 in row 3, column 2 indicates that 3 is related to 2):

	1	2	3	4
1	[1	0	0	0]
2	[0	1	1	0]
3	[0	1	1	0]
4	[0	0	0	1]

(a) Is S reflexive? Justify your answer using the matrix. **(1 mark)**

(b) Is S symmetric? Justify your answer using the matrix. **(1 mark)**

(c) Is S transitive? Verify by checking all relevant pairs. **(2 marks)**